Section 3.3: Basic Rules of Probability

Finite Mathematics MTH 107 Fall 2014

Dr. James Ashe

Mars Hill University

Wednesday 8th October, 2014 & Friday 10th October, 2014

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Review: 3.2

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Probability of an event, E

The *probability* of event E, denoted $p(E)$, is

$$p(E) = \frac{n(E)}{n(S)} \quad \textcolor{red}{\leftarrow \text{number of ways } E \text{ can happen}}$$

if the experiment's outcomes are equally likely.

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Problem 3.3.1. Find the following:

A card is drawn from a 52 card deck. What is the probability of getting a queen of hearts?

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↑ number of ways E can happen

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A single six-sided dice is rolled.
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b. E_2: a number between 1 and 6 (inclusive) is rolled.

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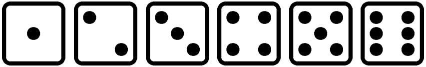

$S = \{1,2,3,4,5,6\}$

a. $p(E_1) = \frac{n(E_1)}{n(S)} = \frac{0}{6} = 0$

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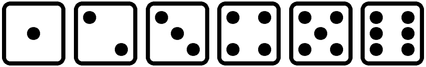

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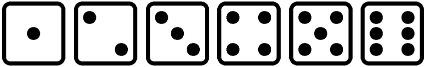

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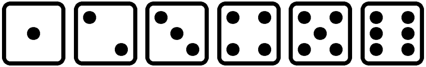

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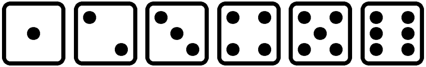

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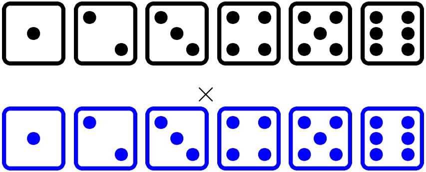

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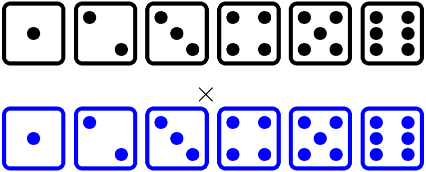

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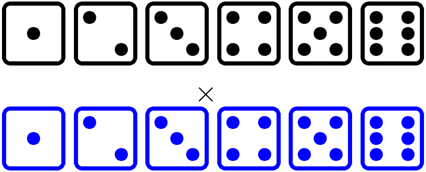

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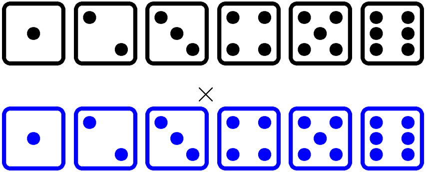

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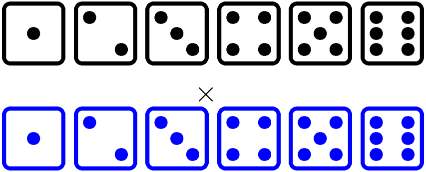

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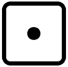

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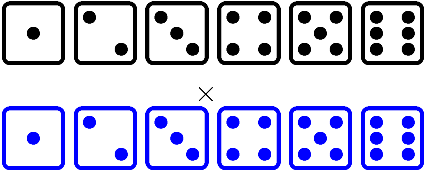

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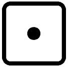 $+$  $= 7$

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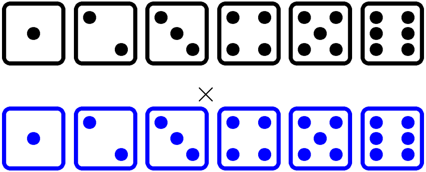

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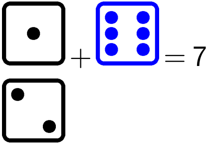

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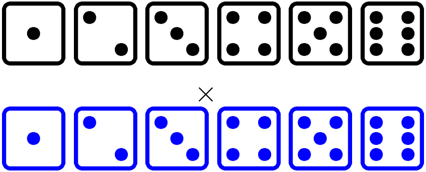

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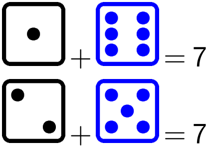

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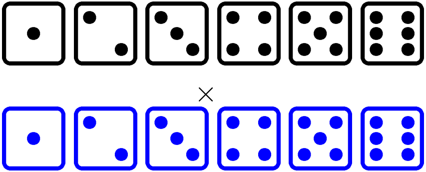

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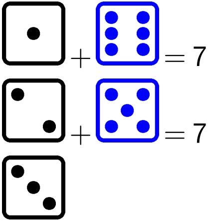

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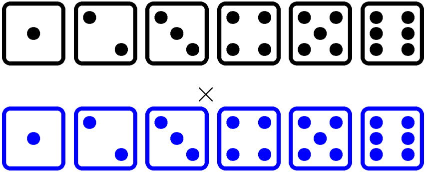

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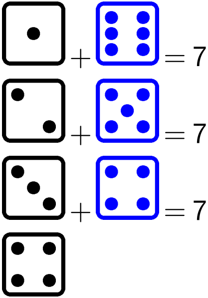

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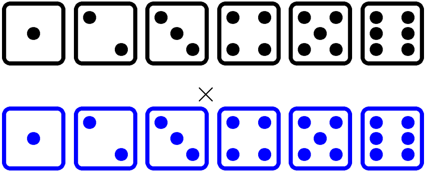

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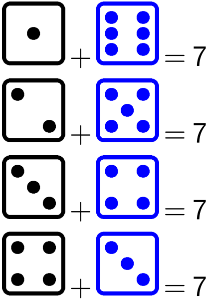

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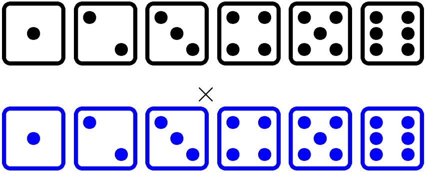

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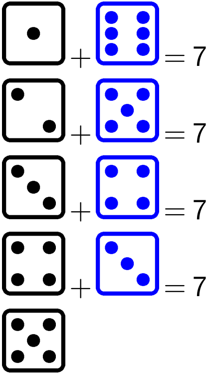

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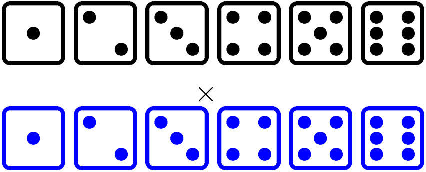

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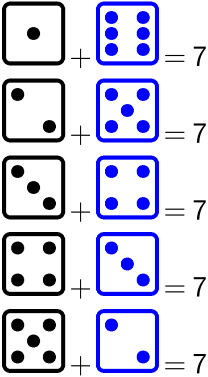

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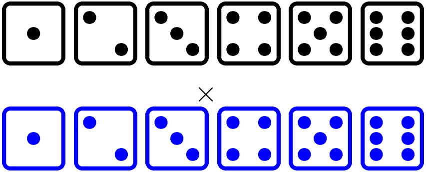

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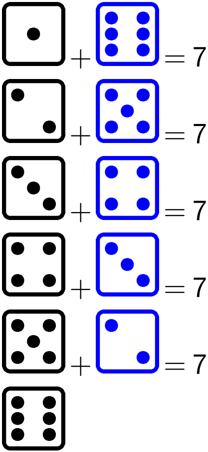

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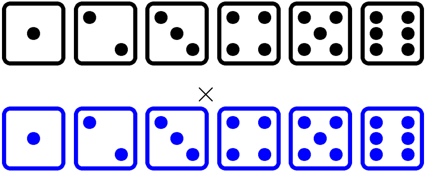
A pair of dice is rolled. Find the probability of:

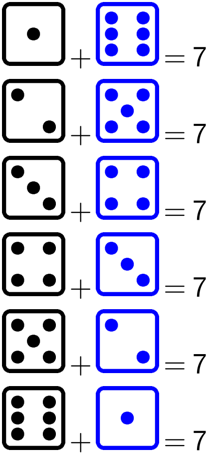
a. E_1: a 7 is rolled



$= 36$ possibilities

b. Need to find:

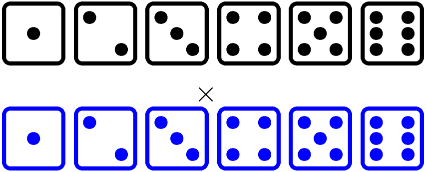
$$p(E_1) = \frac{n(E_1)}{36} \quad \begin{array}{l} \leftarrow \text{ways } E_1 \text{ can occur} \\ \leftarrow \text{total outcomes} \end{array}$$



Example 3.3.2.

A pair of dice is rolled. Find the probability of:

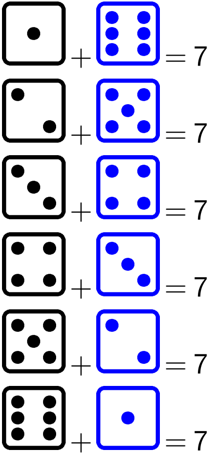
a. E_1: a 7 is rolled



$= 36$ possibilities

b. Need to find:

$$p(E_1) = \frac{6}{36} = \frac{1}{6}$$



Pairs of Dice

Pairs of Dice

Pairs of Dice

Example 3.3.3.

A pair of six-sided dice is rolled. Find the probability of:

b. E_2: a 7 or greater is rolled

Pairs of Dice

Example 3.3.3.

A pair of six-sided dice is rolled. Find the probability of:

b. E_2: a 7 or greater is rolled

a. $p(E_2) = \frac{n(E_2)}{n(S)}$

$= \frac{n(E_2)}{36}$

Pairs of Dice

Example 3.3.3.

A pair of six-sided dice is rolled. Find the probability of:

b. E_2: a 7 or greater is rolled

a. $p(E_2) = \frac{n(E_2)}{n(S)}$

$= \frac{n(E_2)}{36}$

	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)
7	**8**	**9**	**10**	**11**	**12**	

Example 3.3.3.

A pair of six-sided dice is rolled. Find the probability of:

b. E_2: a 7 or greater is rolled

a. $$p(E_2) = \frac{n(E_2)}{n(S)} = \frac{n(E_2)}{36} = \frac{21}{36} = \frac{7}{12}$$

	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)
7	**8**	**9**	**10**	**11**	**12**	

Pairs of Dice

Pairs of Dice

Pairs of Dice

Example 3.3.4.

A pair of six-sided dice is rolled. Find the probability of:

c. E_3: Less than a 5.

Pairs of Dice

Example 3.3.4.

A pair of six-sided dice is rolled. Find the probability of:

c. E_3: Less than a 5.

a. $p(E_3) = \frac{n(E_3)}{n(S)}$

$= \frac{n(E_3)}{36}$

Example 3.3.4.

A pair of six-sided dice is rolled. Find the probability of:

c. E_3: Less than a 5.

a. $p(E_3) = \frac{n(E_3)}{n(S)}$

$= \frac{n(E_3)}{36}$

	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)
7	**8**	**9**	**10**	**11**	**12**	

Pairs of Dice

Example 3.3.4.

A pair of six-sided dice is rolled. Find the probability of:

c. E_3: Less than a 5.

a. $p(E_3) = \frac{n(E_3)}{n(S)}$
$= \frac{n(E_3)}{36}$
$= \frac{6}{36} = \frac{1}{6}$

	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)
7	8	9	10	11	12	

Example 3.3.4.

A pair of six-sided dice is rolled. Find the probability of:

c. E_3: Less than a 5.

a. $$p(E_3) = \frac{n(E_3)}{n(S)} = \frac{n(E_3)}{36} = \frac{6}{36} = \frac{1}{6}$$

Note: Not, $\frac{3}{12}$.

	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)
7	**8**	**9**	**10**	**11**	**12**	

Pairs of Dice

Example 3.3.4.

A pair of six-sided dice is rolled. Find the probability of:

c. E_3: Less than a 5.

a. $p(E_3) = \frac{n(E_3)}{n(S)}$
$= \frac{n(E_3)}{36}$
$= \frac{6}{36} = \frac{1}{6}$

Note: Not, $\frac{3}{12}$.

Also note,

	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
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5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)
7	**8**	**9**	**10**	**11**	**12**	

Pairs of Dice

Example 3.3.4.

A pair of six-sided dice is rolled. Find the probability of:

c. E_3: Less than a 5.

a. $p(E_3) = \frac{n(E_3)}{n(S)}$

$= \frac{n(E_3)}{36}$

$= \frac{6}{36} = \frac{1}{6}$

Note: Not, $\frac{3}{12}$.

Also note,

$P(\text{rolling } 2) = \frac{1}{36}$

$P(\text{rolling } 3) = \frac{2}{36}$

$p(\text{rolling } 4) = \frac{3}{36}$

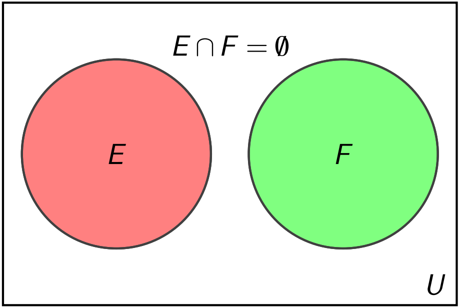
$\frac{1}{36} + \frac{2}{36} + \frac{3}{36} = \frac{6}{36}$

	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)
7	**8**	**9**	**10**	**11**	**12**	

Mutually Exclusive Events

Mutually Exclusive Events

To events that cannot occur at the same time are called *mutually exclusive*.

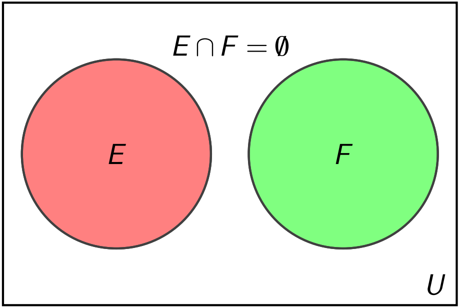


Mutually Exclusive Events

Mutually Exclusive Events

To events that cannot occur at the same time are called *mutually exclusive*.

i.e., E and F are mutually exclusive <u>if and only if</u> $E \cap F = \emptyset$

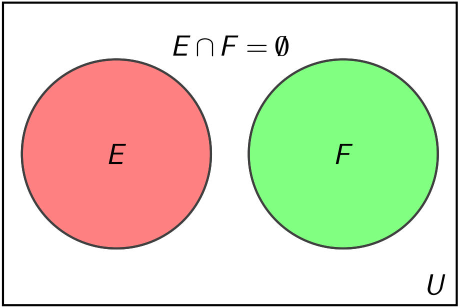


Mutually Exclusive Events

To events that cannot occur at the same time are called *mutually exclusive*.

i.e., E and F are mutually exclusive if and only if $E \cap F = \emptyset$

- Recall: $E \cap F$ is the set of elements in both E and F.



Mutually Exclusive Events

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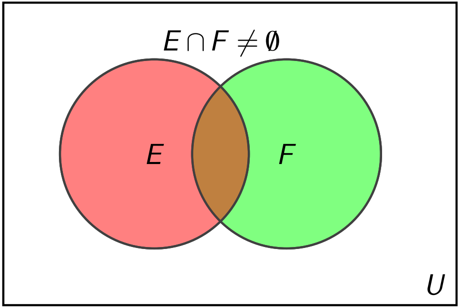
To events that cannot occur at the same time are called *mutually exclusive*.

i.e., E and F are mutually exclusive <u>if and only if</u> $E \cap F = \emptyset$

- Recall: $E \cap F$ is the set of elements in <u>both</u> E <u>and</u> F.

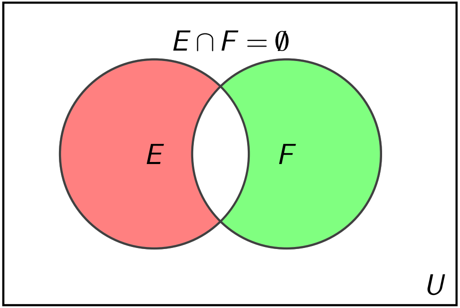


Mutually Exclusive Events

To events that cannot occur at the same time are called *mutually exclusive*.

i.e., E and F are mutually exclusive if and only if $E \cap F = \emptyset$

- Recall: $E \cap F$ is the set of elements in both E and F.



Mutually Exclusive Events

Mutually Exclusive Events

To events that cannot occur at the same time are called *mutually exclusive*.

i.e., E and F are mutually exclusive if and only if $E \cap F = \emptyset$

- Recall: $E \cap F$ is the set of elements in both E and F.

Problem 3.3.3. Determine whether the two events are mutually exclusive

E_1: Being a doctor.

E_2: Being a woman.

Mutually Exclusive Events

Mutually Exclusive Events

To events that cannot occur at the same time are called *mutually exclusive*.

i.e., E and F are mutually exclusive if and only if $E \cap F = \emptyset$

- Recall: $E \cap F$ is the set of elements in both E and F.

Problem 3.3.4. Determine whether the two events are mutually exclusive

E_1: Being alive.

E_2: Being dead.

Mutually Exclusive Events

Mutually Exclusive Events

To events that cannot occur at the same time are called *mutually exclusive*.

i.e., E and F are mutually exclusive if and only if $E \cap F = \emptyset$

- Recall: $E \cap F$ is the set of elements in both E and F.

Problem 3.3.5. Determine whether the two events are mutually exclusive

♡: Drawing a Heart (from a standard 52 card deck).

♠: Drawing a Spade.

Mutually Exclusive Events

Mutually Exclusive Events

To events that cannot occur at the same time are called *mutually exclusive*.

i.e., E and F are mutually exclusive <u>if and only if</u> $E \cap F = \emptyset$

- Recall: $E \cap F$ is the set of elements in <u>both</u> E <u>and</u> F.

Problem 3.3.6. Determine whether the two events are mutually exclusive

J: Drawing a Jack (from a standard 52 card deck).

♠: Drawing a spade.

Probability Rules

Probability Rules

Probability Rules

Example 3.3.5.

What is the probability of drawing a queen OR heart?

$P(Q \cup H) = \frac{n(Q \cup H)}{n(S)}$

Probability Rules

Example 3.3.5.

What is the probability of drawing a queen OR heart?

$$P(Q \cup H) = \frac{n(Q \cup H)}{n(S)} = \frac{n(Q \cup H)}{52}$$

Probability Rules

Example 3.3.5.

What is the probability of drawing a queen OR heart?

$P(Q \cup H) = \frac{n(Q \cup H)}{n(S)} = \frac{n(Q \cup H)}{52}$

$n(Q \cup H) = n(Q) + n(H) - n(Q \cap H) \leftarrow$ **Union/Intersection rule**

Probability Rules

Example 3.3.5.

What is the probability of drawing a queen OR heart?

$$P(Q \cup H) = \frac{n(Q \cup H)}{n(S)} = \frac{n(Q \cup H)}{52}$$

$n(Q \cup H) = n(Q) + n(H) - n(Q \cap H) \leftarrow$ **Union/Intersection rule**

$$Q = \{q\clubsuit, q\spadesuit, q\heartsuit, q\diamondsuit\}$$

$$H = \{2\heartsuit, 3\heartsuit, 4\heartsuit, 5\heartsuit, 6\heartsuit, 7\heartsuit, 8\heartsuit, 9\heartsuit, 10\heartsuit, j\heartsuit, q\heartsuit, k\heartsuit, a\heartsuit\}$$

Probability Rules

Example 3.3.5.

What is the probability of drawing a queen OR heart?

$$P(Q \cup H) = \frac{n(Q \cup H)}{n(S)} = \frac{n(Q \cup H)}{52}$$

$n(Q \cup H) = n(Q) + n(H) - n(Q \cap H) \leftarrow$ **Union/Intersection rule**

$n(Q \cup H) = 4 + 13$

$$Q = \{q\clubsuit, q\spadesuit, q\heartsuit, q\diamondsuit\}$$

$$H = \{2\heartsuit, 3\heartsuit, 4\heartsuit, 5\heartsuit, 6\heartsuit, 7\heartsuit, 8\heartsuit, 9\heartsuit, 10\heartsuit, j\heartsuit, q\heartsuit, k\heartsuit, a\heartsuit\}$$

Probability Rules

Example 3.3.5.

What is the probability of drawing a queen OR heart?

$$P(Q \cup H) = \frac{n(Q \cup H)}{n(S)} = \frac{n(Q \cup H)}{52}$$

$n(Q \cup H) = n(Q) + n(H) - n(Q \cap H) \leftarrow$ **Union/Intersection rule**

$n(Q \cup H) = 4 + 13 - 1$

$$Q = \{q\clubsuit, q\spadesuit, q\heartsuit, q\diamondsuit\}$$

$$H = \{2\heartsuit, 3\heartsuit, 4\heartsuit, 5\heartsuit, 6\heartsuit, 7\heartsuit, 8\heartsuit, 9\heartsuit, 10\heartsuit, j\heartsuit, q\heartsuit, k\heartsuit, a\heartsuit\}$$

Probability Rules

Example 3.3.5.

What is the probability of drawing a queen OR heart?

$$P(Q \cup H) = \frac{n(Q \cup H)}{n(S)} = \frac{n(Q \cup H)}{52} = \frac{16}{52} = \frac{4}{13}$$

$n(Q \cup H) = n(Q) + n(H) - n(Q \cap H) \leftarrow$ **Union/Intersection rule**

$n(Q \cup H) = 4 + 13 - 1$

$$Q = \{q\clubsuit, q\spadesuit, q\heartsuit, q\diamondsuit\}$$

$$H = \{2\heartsuit, 3\heartsuit, 4\heartsuit, 5\heartsuit, 6\heartsuit, 7\heartsuit, 8\heartsuit, 9\heartsuit, 10\heartsuit, j\heartsuit, q\heartsuit, k\heartsuit, a\heartsuit\}$$

Probability Rules

1. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

Example 3.3.5.

What is the probability of drawing a queen OR heart?

$P(Q \cup H) = \frac{n(Q \cup H)}{n(S)} = \frac{n(Q \cup H)}{52} = \frac{16}{52} = \frac{4}{13}$

$n(Q \cup H) = n(Q) + n(H) - n(Q \cap H) \leftarrow$ **Union/Intersection rule**

$n(Q \cup H) = 4 + 13 - 1$

$$Q = \{q\clubsuit, q\spadesuit, q\heartsuit, q\diamondsuit\}$$

$$H = \{2\heartsuit, 3\heartsuit, 4\heartsuit, 5\heartsuit, 6\heartsuit, 7\heartsuit, 8\heartsuit, 9\heartsuit, 10\heartsuit, j\heartsuit, q\heartsuit, k\heartsuit, a\heartsuit\}$$

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.

Example 3.3.5.

What is the probability of drawing a queen OR heart?

$$P(Q \cup H) = \frac{n(Q \cup H)}{n(S)} = \frac{n(Q \cup H)}{52} = \frac{16}{52} = \frac{4}{13}$$

$n(Q \cup H) = n(Q) + n(H) - n(Q \cap H) \leftarrow$ **Union/Intersection rule**

$n(Q \cup H) = 4 + 13 - 1$

$$Q = \{q\clubsuit, q\spadesuit, q\heartsuit, q\diamondsuit\}$$

$$H = \{2\heartsuit, 3\heartsuit, 4\heartsuit, 5\heartsuit, 6\heartsuit, 7\heartsuit, 8\heartsuit, 9\heartsuit, 10\heartsuit, j\heartsuit, q\heartsuit, k\heartsuit, a\heartsuit\}$$

Probability Rules

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.

Example 3.3.5.

What is the probability of drawing a queen OR ace?

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.

Example 3.3.5.

What is the probability of drawing a queen OR ace?

$$P(Q \cup A) = \frac{n(Q \cup A)}{n(S)}$$

Probability Rules

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.

Example 3.3.5.

What is the probability of drawing a queen OR ace?

$$P(Q \cup A) = \frac{n(Q \cup A)}{n(S)} = \frac{n(Q \cup A)}{52}$$

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.

Example 3.3.5.

What is the probability of drawing a queen OR ace?

$P(Q \cup A) = \frac{n(Q \cup A)}{n(S)} = \frac{n(Q \cup A)}{52}$
$n(Q \cup A) = n(Q) + n(A) - n(Q \cap A)$

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.

Example 3.3.5.

What is the probability of drawing a queen OR ace?

$$P(Q \cup A) = \frac{n(Q \cup A)}{n(S)} = \frac{n(Q \cup A)}{52}$$

$$n(Q \cup A) = n(Q) + n(A) - n(Q \cap A)$$

$$Q = \{q\clubsuit, q\spadesuit, q\heartsuit, q\diamondsuit\}$$

$$A = \{a\clubsuit, a\spadesuit, a\heartsuit, a\diamondsuit\}$$

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.

Example 3.3.5.

What is the probability of drawing a queen OR ace?

$$P(Q \cup A) = \frac{n(Q \cup A)}{n(S)} = \frac{n(Q \cup A)}{52}$$

$$n(Q \cup A) = n(Q) + n(A) - n(Q \cap A)$$

$$n(Q \cup A) = 4 + 4$$

$$Q = \{q\clubsuit, q\spadesuit, q\heartsuit, q\diamondsuit\}$$

$$A = \{a\clubsuit, a\spadesuit, a\heartsuit, a\diamondsuit\}$$

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.

Example 3.3.5.

What is the probability of drawing a queen OR ace?

$$P(Q \cup A) = \frac{n(Q \cup A)}{n(S)} = \frac{n(Q \cup A)}{52}$$
$$n(Q \cup A) = n(Q) + n(A) - n(Q \cap A)$$
$$n(Q \cup A) = 4 + 4 - 0 \leftarrow Q \cap A = \emptyset$$
$$Q = \{q\clubsuit, q\spadesuit, q\heartsuit, q\diamondsuit\}$$
$$A = \{a\clubsuit, a\spadesuit, a\heartsuit, a\diamondsuit\}$$

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.

Example 3.3.5.

What is the probability of drawing a queen OR ace?

$$P(Q \cup A) = \frac{n(Q \cup A)}{n(S)} = \frac{n(Q \cup A)}{52} = \frac{8}{52}$$

$$n(Q \cup A) = n(Q) + n(A) - n(Q \cap A)$$

$$n(Q \cup A) = 4 + 4 - 0 \leftarrow Q \cap A = \emptyset$$

$$Q = \{q\clubsuit, q\spadesuit, q\heartsuit, q\diamondsuit\}$$

$$A = \{a\clubsuit, a\spadesuit, a\heartsuit, a\diamondsuit\}$$

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.

Example 3.3.5.

What is the probability of drawing a queen OR ace?

$P(Q \cup A) = \frac{n(Q \cup A)}{n(S)} = \frac{n(Q \cup A)}{52} = \frac{8}{52}$

$n(Q \cup A) = n(Q) + n(A) - n(Q \cap A)$

$n(Q \cup A) = 4 + 4 - 0 \leftarrow Q \cap A = \emptyset$

$Q = \{q\clubsuit, q\spadesuit, q\heartsuit, q\diamondsuit\}$

$A = \{a\clubsuit, a\spadesuit, a\heartsuit, a\diamondsuit\}$

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.
2. $n(A \cup B) = n(A) + n(B)$ if $A \cap B = \emptyset$.

Example 3.3.5.

What is the probability of drawing a queen OR ace?

$$P(Q \cup A) = \frac{n(Q \cup A)}{n(S)} = \frac{n(Q \cup A)}{52} = \frac{8}{52}$$
$$n(Q \cup A) = n(Q) + n(A) - n(Q \cap A)$$
$$n(Q \cup A) = 4 + 4 - 0 \leftarrow Q \cap A = \emptyset$$
$$Q = \{q\clubsuit, q\spadesuit, q\heartsuit, q\diamondsuit\}$$
$$A = \{a\clubsuit, a\spadesuit, a\heartsuit, a\diamondsuit\}$$

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.
2. $p(A \cup B) = p(A) + p(B)$ if $A \cap B = \emptyset$.

Example 3.3.5.

What is the probability of drawing a queen OR ace?

$$P(Q \cup A) = \frac{n(Q \cup A)}{n(S)} = \frac{n(Q \cup A)}{52} = \frac{8}{52}$$
$$n(Q \cup A) = n(Q) + n(A) - n(Q \cap A)$$
$$n(Q \cup A) = 4 + 4 - 0 \leftarrow Q \cap A = \emptyset$$
$$Q = \{q\clubsuit, q\spadesuit, q\heartsuit, q\diamondsuit\}$$
$$A = \{a\clubsuit, a\spadesuit, a\heartsuit, a\diamondsuit\}$$

Probability Rules

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.
2. $p(A \cup B) = p(A) + p(B)$ if $A \cap B = \emptyset$.

Example 3.3.5.

What is the probability of NOT drawing a queen or ace?

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.
2. $p(A \cup B) = p(A) + p(B)$ if $A \cap B = \emptyset$.

Example 3.3.5.

What is the probability of NOT drawing a queen or ace?

$$P((Q \cup A)') = \frac{n((Q \cup A)')}{n(S)} = \frac{n((Q \cup A)')}{52}$$

Probability Rules

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.
2. $p(A \cup B) = p(A) + p(B)$ if $A \cap B = \emptyset$.

Example 3.3.5.

What is the probability of NOT drawing a queen or ace?

$$P((Q \cup A)') = \frac{n((Q \cup A)')}{n(S)} = \frac{n((Q \cup A)')}{52} = \frac{44}{52}$$

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.
2. $p(A \cup B) = p(A) + p(B)$ if $A \cap B = \emptyset$.

Example 3.3.5.

What is the probability of NOT drawing a queen or ace?

$P((Q \cup A)') = \frac{n((Q \cup A)')}{n(S)} = \frac{n((Q \cup A)')}{52} = \frac{44}{52}$

$n(Q \cup A) = 8$

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.
2. $p(A \cup B) = p(A) + p(B)$ if $A \cap B = \emptyset$.

Example 3.3.5.

What is the probability of NOT drawing a queen or ace?

$P((Q \cup A)') = \frac{n((Q \cup A)')}{n(S)} = \frac{n((Q \cup A)')}{52} = \frac{44}{52}$
$n(Q \cup A) = 8$
$n((Q \cup H)) = 52 - 8$

Probability Rules

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.
2. $p(A \cup B) = p(A) + p(B)$ if $A \cap B = \emptyset$.
3.

Example 3.3.5.

What is the probability of NOT drawing a queen or ace?

$P((Q \cup A)') = \frac{n((Q \cup A)')}{n(S)} = \frac{n((Q \cup A)')}{52} = \frac{44}{52}$
$n(Q \cup A) = 8$
$n((Q \cup H)) = 52 - 8 = 44$

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.
2. $p(A \cup B) = p(A) + p(B)$ if $A \cap B = \emptyset$.
3. $n(A') = n(U) - n(A)$.

Example 3.3.5.

What is the probability of NOT drawing a queen or ace?

$P((Q \cup A)') = \frac{n((Q \cup A)')}{n(S)} = \frac{n((Q \cup A)')}{52} = \frac{44}{52}$
$n(Q \cup A) = 8$
$n((Q \cup H)) = 52 - 8 = 44$

Probability Rules

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.
2. $p(A \cup B) = p(A) + p(B)$ if $A \cap B = \emptyset$.
3. $p(A') = \frac{n(S)}{n(S)} - p(A)$.

Example 3.3.5.

What is the probability of NOT drawing a queen or ace?

$P((Q \cup A)') = \frac{n((Q \cup A)')}{n(S)} = \frac{n((Q \cup A)')}{52} = \frac{44}{52}$
$n(Q \cup A) = 8$
$n((Q \cup H)) = 52 - 8 = 44$

Probability Rules

1. $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.
2. $p(A \cup B) = p(A) + p(B)$ if $A \cap B = \emptyset$.
3. $p(A') = 1 - p(A)$.

Example 3.3.5.

What is the probability of NOT drawing a queen or ace?

$P((Q \cup A)') = \frac{n((Q \cup A)')}{n(S)} = \frac{n((Q \cup A)')}{52} = \frac{44}{52}$

$n(Q \cup A) = 8$

$n((Q \cup H)) = 52 - 8 = 44$

Problems

	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)
7	**8**	**9**	**10**	**11**	**12**	

Example 3.3.6.

A pair of six-sided dice is rolled. Find the probability of rolling:

a. E_3: Less than a 5.

b. E_4: 5 or greater.

	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)
7	**8**	**9**	**10**	**11**	**12**	

Problems

Example 3.3.6.

A pair of six-sided dice is rolled.
Find the probability of rolling:

a. E_3: Less than a 5.

b. E_4: 5 or greater.

c. E_5: 5 or 4.

	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)
7	**8**	**9**	**10**	**11**	**12**	

Problems

Example 3.3.6.

A pair of six-sided dice is rolled. Find the probability of rolling:

a. E_3: Less than a 5.

b. E_4: 5 or greater.

c. E_5: 5 or 4.

d. E_6: Odd or 5.

	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)
7	**8**	**9**	**10**	**11**	**12**	

Problem 3.3.7. A card is randomly dealt from a standard 52 card deck. Find the probability that the card is as stated.

Problem 3.3.8. A card is randomly dealt from a standard 52 card deck. Find the probability that the card is as stated.

a. A 5 and a black.

Problems

Problem 3.3.9. Find the relative frequency with which a customer's bill is as stated.

Problems

Problem 3.3.10. Find the relative frequency with which a customer's bill is as stated.

a. Between \$40.00 and \$79.99.

Problems

Size of Bill	# of Customers
below $20	200
$20-39.99	100
$40-59.99	150
$60-79.99	50
$80-99.99	300
above $100	100

Probabilities and Venn Diagrams

Probability that a student does NOT fail Math: $1 - .65$

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

Probability that a student does NOT fail Math: $1 - .65 = .35$

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.

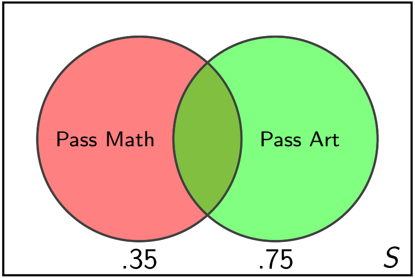
Probabilities and Venn Diagrams

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.

Prob. Passes at least 1



Probability that a student passes Math AND Art is: $P(M \cap A)$

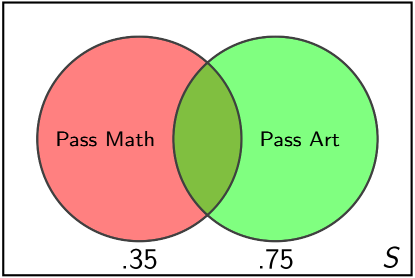
Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.

b. passes both.

Prob. Passes at least 1



Probability that a student passes Math AND Art is: $P(M \cap A)$

Pass both↓

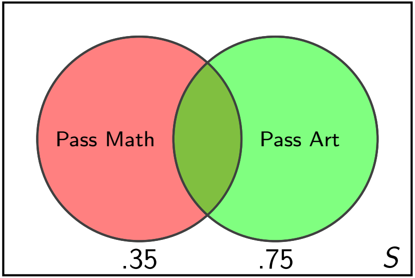
$$\begin{array}{ccccccc} P(M \cup A) & = & P(M) & + & P(A) & - & P(M \cap A) \\ .85 & = & & & & & \end{array}$$

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.

Prob. Passes at least 1



Probability that a student passes Math AND Art is: $P(M \cap A)$

Pass both↓

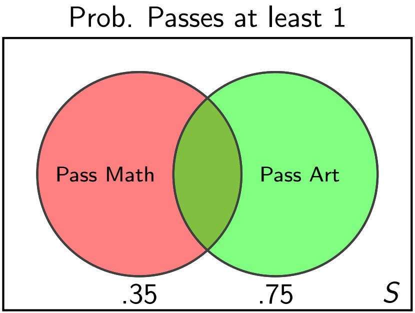
$$\begin{array}{ccccccc} P(M \cup A) & = & P(M) & + & P(A) & - & P(M \cap A) \\ .85 & = & .35 & + & & & \end{array}$$

Probabilities and Venn Diagrams

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.



Probability that a student passes Math AND Art is: $P(M \cap A)$

Pass both↓

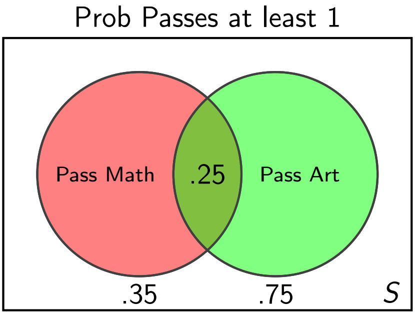
$$
\begin{array}{ccccccccc}
P(M \cup A) & = & P(M) & + & P(A) & - & P(M \cap A) \\
.85 & = & .35 & + & .75 & - &
\end{array}
$$

Probabilities and Venn Diagrams

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.



Probability that a student passes Math AND Art is: $P(M \cap A)$

Pass both↓

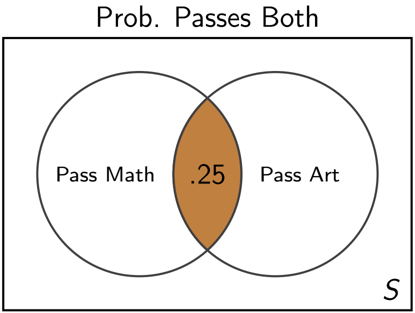
$$\begin{array}{ccccccccc} P(M \cup A) & = & P(M) & + & P(A) & - & P(M \cap A) \\ .85 & = & .35 & + & .75 & - & .25 \end{array}$$

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.

b. passes both.



Pass both↓

$$\begin{array}{ccccccccc} P(M \cup A) & = & P(M) & + & P(A) & - & P(M \cap A) \\ .85 & = & .35 & + & .75 & - & .25 \end{array}$$

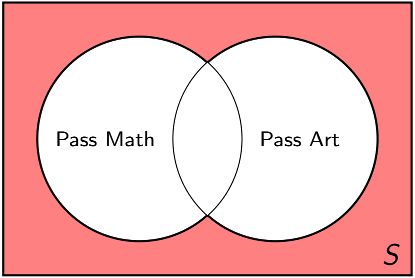
Probabilities and Venn Diagrams

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.
c. fails both.

Prob. Fails Both

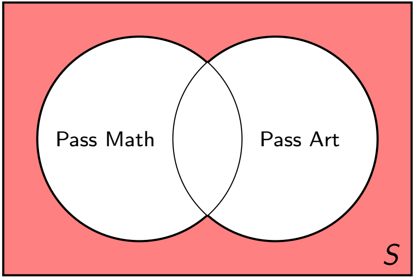


Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.
c. fails both.

Prob. Fails Both



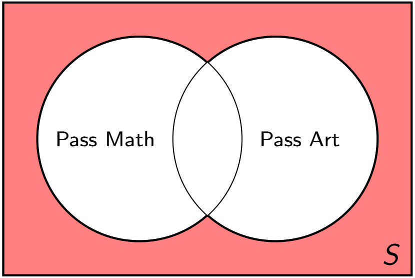
Probability that a student does NOT pass at least one.

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.
c. fails both.

Prob. Fails Both



Probability that a student does NOT pass at least one.

$1 - .85$

Probabilities and Venn Diagrams

Example 3.3.7.

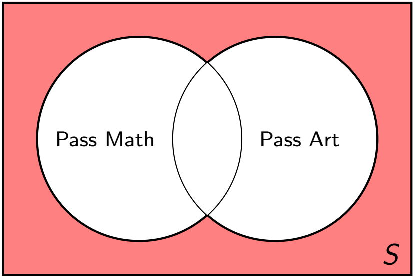
The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.

b. passes both.

c. fails both.

Prob. Fails Both



Probability that a student does NOT pass at least one.

$$1 - .85 = .15$$

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.
c. fails both.
d. passes only one.

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.
c. fails both.
d. passes only one.

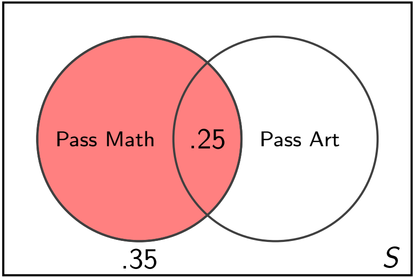
Probability that a student passes just Math

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.
c. fails both.
d. passes only one.

Prob. Passes Math$= .35$



Probability that a student passes just Math

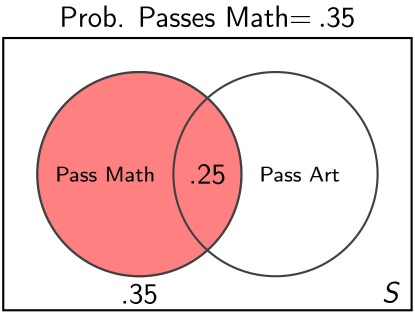
$P(M) \quad - \quad P(M \cap A)$

Probabilities and Venn Diagrams

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.
c. fails both.
d. passes only one.



Probability that a student passes just Math

$$P(M) \quad - \quad P(M \cap A)$$
$$.35 \quad - \quad .25$$

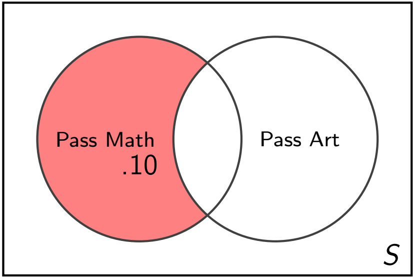
Probabilities and Venn Diagrams

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.
c. fails both.
d. passes only one.

Prob. Passes Only Math



Probability that a student passes just Math

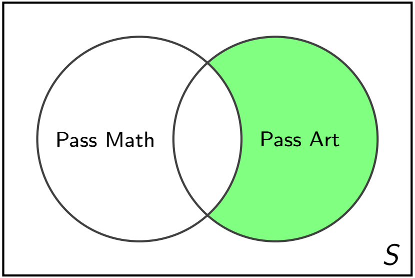
$$\begin{array}{ccccc} P(M) & - & P(M \cap A) & & \\ .35 & - & .25 & & = .10 \end{array}$$

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.
c. fails both.
d. passes only one.

Prob. Passes Only Art



Probability that a student passes just Math OR just Art

$P(A) \quad - \quad P(M \cap A)$

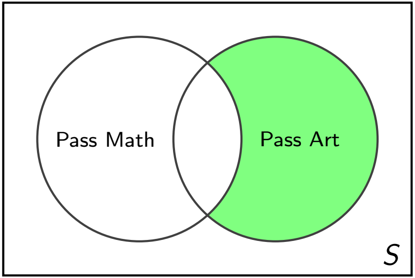
Probabilities and Venn Diagrams

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.
c. fails both.
d. passes only one.

Prob. Passes Only Art



Probability that a student passes just Math OR just Art

$$P(A) \quad - \quad P(M \cap A)$$
$$.75 \quad - \quad .25 \quad = .5$$

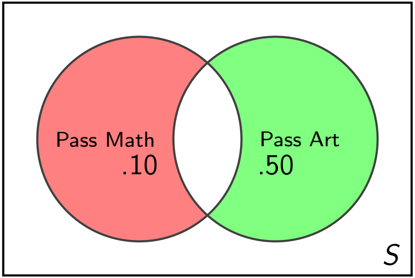
Probabilities and Venn Diagrams

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.
c. fails both.
d. passes only one.

Prob. Passes only Math or Art



Probability that a student passes just Math OR just Art

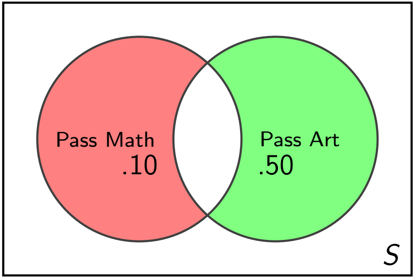
$$\begin{array}{ccccc} P(A) & - & P(M \cap A) & & \\ .75 & - & .25 & = .5 \end{array}$$

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.
c. fails both.
d. passes only one.

Prob. Passes only Math or Art



Probability that a student passes just Math OR just Art

$.10 + .50$

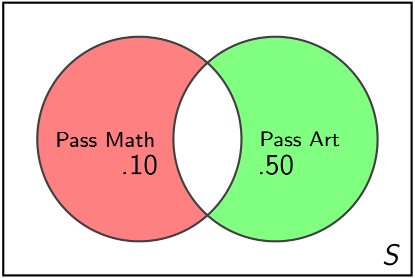
Probabilities and Venn Diagrams

Example 3.3.7.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

a. passes Math.
b. passes both.
c. fails both.
d. passes only one.

Prob. Passes only Math or Art



Probability that a student passes just Math OR just Art

$.10 + .50 = .60$